DA6360 : WS1

1.(a) $f(x) = \max\{|x-a|, |x-b|\}$
↳ larger of the 2 distances from x to pts. a & b.

⇒ $x \leq a$
$x - a \leq 0$ & $x - b \leq 0$
$|x-a| = a - x$, $|x-b| = b - x$
$a < b$ ⇒ $b - x > a - x$
$f(x) = b - x$ always

⇒ $a \leq x \leq b$
$x - a \geq 0$, $x - b \leq 0$
$|x-a| = x - a$, $|x-b| = b - x$
$f(x) = \max(x-a, b-x)$

⇒ $x \geq b$
$f(x) = x - a$

⇒ for $x \leq a$, $f(x) = b - x$ ↓ as x ↑
$f(a) = \max(0, b-a) = b - a$

⇒ for $x \geq b$, $f(x) = x - a$ ↑ as x ↑
$f(b) = b - a$

$f(x) = \max(x-a, b-a)$
$x - a = b - x$
$2x - a = b$
$2x = a + b$ ⇒ $x = \frac{a+b}{2}$

(ii) minimiser → pt. at which min. is attained

$x_* = \arg\min_{x \in R} f(x) = \frac{a+b}{2}$

(i) $f(x) = \max(x-a, b-a)$

$$f\left(\frac{a+b}{2}\right) = \max\left(\frac{a+b}{2} - a, b-a\right)$$

$$= \max\left(\frac{a+b-2a}{2}, b-a\right)$$

$$= \max\left(\frac{b-a}{2}, b-a\right)$$

$$f* = \min_{x \in R} f(x) = \frac{b-a}{2}$$

(iii) Yes, the minimiser is unique -

$x < \frac{a+b}{2}$, $f(x) = b-x \downarrow$, as $x \uparrow$

$x > \frac{a+b}{2}$, $f(x) = x-a \uparrow$, as $x \uparrow$

so the func^n decreases upto $(a+b)/2$ & increases after, exactly at pt. we have the minimum.

2. P - ₹20, T - ₹25, O - ₹30.

Total - 100. Buy atleast 2 diff. items.

$x_1, x_2, x_3 \geq 0$

$20x_1 + 25x_2 + 30x_3 \leq 100$

$$y_i = \begin{cases} 1 & \text{if item } i \text{ is bought} \\ 0 & \text{o.w.} \end{cases}$$

$y_1, y_2, y_3 \in \{0, 1\}$ → binary var.

$$y_1 + y_2 + y_3 \geq 2$$

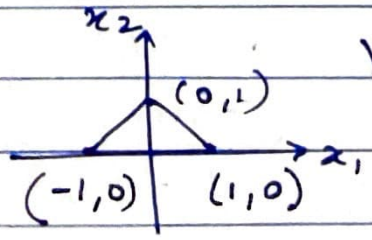
with the budget,

$$0 \leq x_1 \leq 5y_1$$
$$0 \leq x_2 \leq 4y_2$$
$$0 \leq x_3 \leq \frac{10}{3} y_3$$

$$\max \; x_1 + x_2 + x_3$$

$$\max_{\substack{x_1, x_2, x_3 \\ y_1, y_2, y_3}} \; x_1 + x_2 + x_3$$
$$\text{s.t. } 20x_1 + 25x_2 + 30x_3 \leq 100$$
$$0 \leq x_1 \leq 5y_1$$
$$0 \leq x_2 \leq 4y_2$$
$$0 \leq x_3 \leq 10/3 \; y_3$$
$$y_1 + y_2 + y_3 \geq 2$$
$$x_1, x_2, x_3 \geq 0$$
$$y_1, y_2, y_3 \in \{0, 1\}$$

3. (1)



$V = \{(-1,0), (1,0), (0,1)\}$

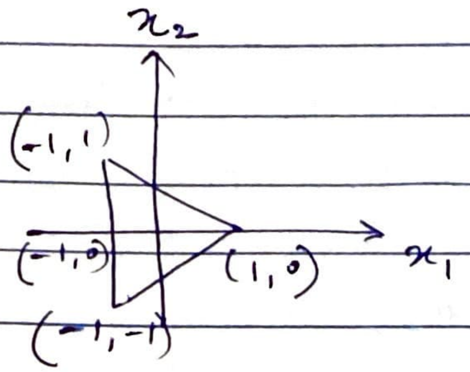
$E \rightarrow$ $x_2 = 0$

$x_2 = x_1 + 1$

$x_2 = -x_1 + 1$

$$\{(x_1, x_2) : x_2 \geq 0, \; x_2 \leq x_1 + 1, \; x_2 \leq -x_1 + 1\}$$

(2)



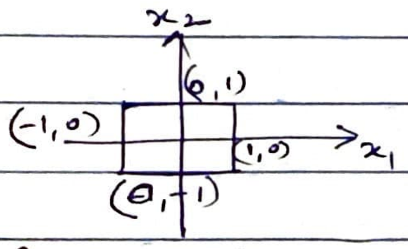
$x_1 = -1$

$x_2 = \frac{1 - x_1}{2}$

$x_2 = \frac{x_1 - 1}{2}$

$$\left\{ (x_1, x_2) : x_1 = -1, \frac{x_1 - 1}{2} \leq x_2 \leq \frac{1 - x_1}{2} \right\}$$

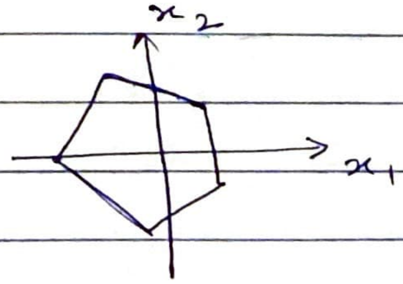
(3)



$(\pm 1, \pm 1)$

$$\left\{ (x_1, x_2) : -1 \leq x_1 \leq 1, \ -1 \leq x_2 \leq 1 \right\}$$

(4)



regular pentagon centered at the origin

$$\left\{ x \in R^2 : A_i^T x \leq b, \ i = 1, 2 \ldots 5 \right\}$$

where, $b = \frac{1}{2} \cot 36°$

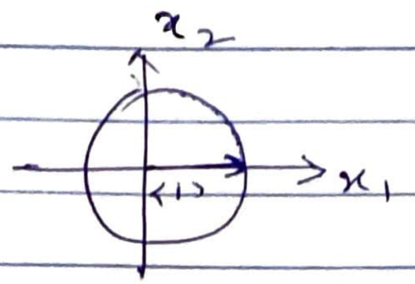
5 normals are

$A_1 = \begin{pmatrix} \sin 18° \\ \cos 18° \end{pmatrix}$, $A_2 = \begin{pmatrix} \cos 36° \\ -\sin 36° \end{pmatrix}$,

$A_3 = \begin{pmatrix} \cos 36° \\ \sin 36° \end{pmatrix}$, $A_4 = \begin{pmatrix} \sin 18° \\ -\cos 18° \end{pmatrix}$,

$A_5 = \begin{pmatrix} \sin 18° \\ -\cos 18° \end{pmatrix}$, $A_5 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

⑤



$\{(x_1, x_2) : x_1^2 + x_2^2 \leq 1\}$

4. we have 5 linear & 5 quad. functions

$\boxed{\text{Fig. 1}}$

1. $x_1 + x_2$

max = 1, maxi. $\overline{BC}$ (1,0) & (0,1) ⇒ not unique

min = −1, mini. (−1,0) ⇒ unique

2. $-x_1 + x_2$

max = 1, maxi. (−1,0) (0,1) ⇒ not unique

min = −1, min. (1,0) ⇒ unique

3. $x_1 - x_2$

max = 1, maxi. (1,0) ⇒ unique

min = −1, mini. (−1,0) (0,1) ⇒ not unique

4. $-x_1 - x_2$

max = 1, maxi. (−1,0) ⇒ unique

min = −1, mini. (0,1) (1,0) → not unique

5. $2x_1 + x_2$

max = 2, maxi. (1,0) → unique

min = −2, mini (−1,0) ⇒ unique

6. $(x_1 - 1)^2 + (x_2 - 2)^2$

min. = 2 ~~mini~~. (0,1) ⇒ unique

$\frac{d' f(x)}{dx} = 2(x_1 - 1)$

max = 8 ⇒ maxi. (−1,0) ⇒ unique